

Track 1 — Parts List & Procurement

Dedicated hardware to acquire and validate the OBD-II + TPMS + GPS data path
Car Automation IoT Internship • For budget sign-off

1. What We Are Buying For

This list covers everything needed to acquire and test the Track 1 vehicle-data path end to end: reading OBD-II parameters in-house, reading our own tyre sensors, tagging everything with GPS, and streaming it to the backend. It is built around the Vehicle Data Acquisition Standard (v1.0.0).

This is a fresh, dedicated set of hardware for Track 1. All core boards (ESP32, GPS, CAN) are purchased new rather than reused, so Track 1 has its own standalone rig that is not shared with or cannibalised by other work. A two-board layout is used: one board for OBD + TPMS acquisition, one for GPS + CAN.

The data path is three separate subsystems, not one feed:

- OBD adapter → ESP32 → drivetrain, engine health and fuel data (RPM, speed, coolant, load, throttle, intake, fuel level, DTCs).
- 433 MHz receiver + our own tyre sensors → the 16-byte TPMS packet (tyre pressure and temperature).
- GPS module → location, timestamp and satellite count.

In-house requirement: the OBD adapter is wired into our own hardware and decoded in our own code — the adapter’s phone app is not used. This is why the wired ELM327 is the primary pick over the app-based Bluetooth/WiFi units, even though those log faster: the app path is not permitted.

2. Core Hardware

The controller boards and core modules, purchased fresh for a dedicated Track 1 rig.

Item	Qty	Unit ₹	Total ₹	Role / Notes
ESP32 development board	2	500–650 ea	1,000–1,300	A spare board protects against a bricked/damaged unit during hardware work.
NEO-6M GPS module	1	350–500	350–500	With ceramic patch antenna. Provides location, UTC timestamp and satellite count for the 32-byte packet.
MCP2515 CAN module	2	150–250 ea	300–500	CAN interface (TJA1050 transceiver). One as the CAN/backup OBD read-path (reading PIDs directly off CAN), one spare / for the second board. Kept as a fallback if the ELM327 proves unreliable.

3. OBD-II Acquisition

Item	Qty	Approx ₹	Notes
ELM327 OBD-II adapter	1	1,300–2,000	PRIMARY pick. Wired, all OBD2 protocols incl. ISO 15765 (CAN). See \$5 on the wired-connection choice (USB vs

Item	Qty	Approx ₹	Notes
			UART pins) — decide before ordering. Buy a decent unit, not the cheapest clone (clone chips are the usual cause of “not working properly”).
OBD-II extension cable	1	400–500	16-pin male-to-female, ~1–1.5 m. Lets us plug in comfortably and work without contorting under the dash while testing.
USB-to-TTL / connection interface	1	200–400	Only if using a USB adapter with the ESP32 — the ESP32 is not a USB host, so a USB ELM327 does not plug straight in. Not needed if a UART-pin ELM327 or a laptop is used. See §5.

4. TPMS Acquisition & Wiring

4.1 TPMS (separate 433 MHz path)

Item	Qty	Approx ₹	Notes
Aftermarket TPMS sensors (set of 4)	1 set	2,000–2,500	External valve-cap type, 433 MHz. Sold as kits with a display we don’t need. IMPORTANT: packet format varies by brand — confirm the brand before ordering, it determines the decode work.
433 MHz receiver module (CC1101)	1	150–300	SPI transceiver, reads the TPMS broadcasts for the ESP32 to decode. Preferred over a plain superheterodyne receiver (SRX882) because it is tunable and better suited to decoding varied TPMS packets.

4.2 Wiring & misc

Item	Qty	Approx ₹	Notes
Jumper wires (M-M, M-F)	1 set	80	Connections between modules and the ESP32 boards.
Small perfboard / breadboard	1	60	Mounting the receiver and power rails cleanly.
Breadboard power / USB cables	1 set	100	Powering and flashing both boards.

5. Decision Before Ordering: How the Adapter Connects

This must be decided before buying the ELM327, because it changes which unit to order. A USB ELM327 has a USB plug, but the ESP32 is not a USB host — a USB dongle does not plug straight into the ESP32. There are three viable setups:

Setup	What it means	What to buy
1. USB adapter → laptop	Simplest to start. Laptop runs our decode code and sends data to the backend over its own internet. Good for the first real-car test — fastest way to prove the read path works. Downside: a laptop sits in the car during testing.	USB ELM327 + laptop
2. UART-pin adapter → ESP32	Fully in-house, no laptop. Requires an ELM327 board that exposes raw TX/RX serial pins (not a sealed USB dongle). Cleanest final setup — the ESP32 reads it directly over UART.	UART-pin ELM327
3. USB adapter → USB-host module → ESP32	Works but adds a USB-host interface and complexity. Only if a USB unit must be used with the ESP32.	USB ELM327 + USB-host module

Recommended approach: start with Setup 1 (USB adapter + laptop) to prove the car actually talks and the decode logic is correct — this is the fastest way to get real data and de-risk the whole track. Then move to Setup 2 (UART-pin adapter → ESP32) for the final in-house version once the decode is proven.

6. Alternatives Considered

6.1 OBD adapter alternatives

Option	Status	Approx ₹	Trade-off
Wired USB ELM327	PRIMARY	1,300–2,000	Wires directly into our hardware; simple UART/serial-style integration; fits the in-house requirement. Basic buffer can lag under fast polling — mitigated by our 1 Hz poll rate.
Vgate iCar Pro (WiFi)	Alternative	2,800–4,200	Phone-free wireless link, higher throughput than cheap clones. But it needs WiFi socket handling in firmware and leans toward app-based logging — less clean for direct in-house ESP32 wiring.
OBDLink MX+ (Bluetooth)	Alternative	6,500–8,000	Best data quality — STN chipset, large CAN buffer, up to 4× sample rate. But it is built for app-based logging (the path ruled out) and needs a Bluetooth stack in firmware. Overkill for our 1 Hz in-house use.

6.2 433 MHz receiver alternatives

Option	Status	Approx ₹	Trade-off
CC1101 module	PRIMARY	150–300	Tunable SPI transceiver, better for decoding varied TPMS packet formats. The safer pick given TPMS decode is the hard part.
SRX882 superheterodyne receiver	Alternative	100–150	Cheaper and simpler, but cruder — fixed, less capable for decoding non-trivial TPMS packets. Fine only if the chosen sensor packet turns out to be simple.

6.3 OBD read method — adapter vs direct CAN

ELM327 adapter (chosen): the adapter handles the low-level protocol and hands us clean hex; we decode PIDs in our own code. This is the requested approach and the fastest to working data.

Direct CAN via MCP2515 (backup): we are also buying the MCP2515 modules, so we can read OBD PIDs straight off the CAN bus without any adapter if needed — arguably the most in-house option of all. Kept as a fallback if the ELM327 proves unreliable.

7. Cost Summary

Section	Approx ₹
Core hardware (2× ESP32, GPS, 2× CAN)	1,650 – 2,300
OBD acquisition (adapter, cable, interface)	1,900 – 2,900
TPMS acquisition (sensors + CC1101)	2,150 – 2,800
Wiring & misc	~240
Track 1 total	≈ 5,900 – 8,200

8. Sourcing

Parts can be bought locally or online:

- Local: Mercy Electronics (Tambaram), Swastik Electronics (near VIT).
- Online: Robu.in, Robocraze, ThinkRobotics, Quartz Components, and Amazon.in for anything not stocked locally.